Exercise list R2c — Financial planning and forecasting for valuation

Worked solutions

Statements are in `Listas/lista-r2c-financial-planning-forecasting.md`; each exercise is cited by number and printed page and is not restated. Short answers: `lista-r2c-financial-planning-forecasting-solutions-short.pdf`. Editions are the local copies: Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011), Ch. 18; Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed., Ch. 3–4; Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed., Cap. 4 ("NAF" = *necessidade de aporte financeiro* = EFN). Amounts carry the units of the statement (dollars, $ thousands or $ millions). Numbers marked "illustration" are not part of the statement.

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1. Cost of goods sold by the percent of sales method — Berk 18.1, p. 558

What is asked. Next year's cost of goods sold (COGS) when sales rise from $100,000 to $110,000, assuming COGS keeps this year's ratio to sales.

Approach. The percent of sales method (Berk §18.2) assumes that an item that moves with activity keeps the same fraction of sales. So we measure the fraction today and apply it to forecast sales. This is the first line of any pro forma income statement.

Notation.

- $S_0 = 100{,}000$, $S_1 = 110{,}000$: sales this year and next year ($).
- $C_0 = 72{,}000$: COGS this year ($); $c = C_0/S_0$: COGS as a fraction of sales; C_1: forecast COGS.

Step 1 — Measure the ratio. We divide this year's COGS by this year's sales:

$$c = \frac{C_0}{S_0} = \frac{72{,}000}{100{,}000} = 0.72. \tag{1}$$

Step 2 — Apply it to forecast sales. The ratio is assumed constant, so

$$C_1 = c\,S_1 = (0.72)(110{,}000) = 79{,}200. \tag{2}$$

Answer . Forecast COGS = \$79,200 (72% of \$110,000).

Intuition. Holding COGS at 72% of sales assumes the gross margin (28%) is stable: each extra dollar of sales brings 72 cents of extra cost. It is a good first guess for a mature business; a firm with fixed production costs would see COGS grow more slowly than sales.

Common mistakes.

- Growing COGS by a different rate than sales (e.g. by last year's COGS growth) — the method ties every proportional item to the *sales* forecast.
- Applying the ratio to this year's sales instead of next year's.

Ref: Berk §18.2, "Percent of Sales Method", pp. 537–538; Example 18.1, p. 539.

2. Reading external financing required — Higgins 3.1 and 3.4, p. 103

What is asked. (3.1) What a *negative* external financing required means. (3.4a) Whether a larger receivables forecast raises external funding required. (3.4b) Whether cash flow forecasts usually show larger needs than pro forma statements. (3.4c) Whether an annual forecast with no external need rules out cash shortfalls during the year.

Approach. In a pro forma balance sheet, external funding required (EFN) is the *plug*: the amount that makes assets equal liabilities plus equity. Every question follows from the identity that defines it and from the date at which the identity is evaluated.

Notation.

- TA: projected total assets; L: projected liabilities (without new financing); E: projected equity.
- $EFN = TA - (L + E)$: external funding required at the forecast date.

Step 1 — The defining identity. By construction,

$$EFN = TA - (L + E). \tag{3}$$

A positive value means projected uses (assets) exceed projected sources: money must be raised.

Step 2 — (3.1) Negative EFN. If $EFN < 0$, sources exceed uses: the company will generate more cash than its plan absorbs. Illustration: projected assets 1,000 with cash at its minimum of 100, and $L + E = 1{,}040$; then $EFN = 1{,}000 - 1{,}040 = -40$. Setting the plug to zero and adding the 40 to cash gives cash $= 100 + 40 = 140$ and assets $= 900 + 140 = 1{,}040 = L + E$: the balance sheet balances.

Answer (3.1). A negative EFN is a projected cash *surplus* above the desired minimum: cash (or debt repayment, dividends, buybacks) absorbs it.

Step 3 — (3.4a) Receivables. Receivables are part of TA in (3). Holding L and E fixed, $\partial EFN/\partial AR = 1$: in the illustration, raising receivables by 50 moves EFN from -40 to $-40+50 = 10$.

Answer (3.4a). True: every extra dollar tied up in receivables must be financed.

Step 4 — (3.4b) Cash flow forecasts. A cash flow forecast and a pro forma balance sheet rearrange the same accounting identities; Higgins' chapter summary states that cash flow forecasts "yield the same need for external funding as a pro forma projection, given the same assumptions."

Answer (3.4b). False: with the same assumptions and no errors, the two estimates are identical.

Step 5 — (3.4c) Intra-year needs. (3) is evaluated on one date (December 31). A seasonal firm can build inventory and receivables in mid-year and need financing then, even if by year-end the cash has come back.

Answer (3.4c). False: an annual forecast says nothing about needs between forecast dates; monthly cash budgets or quarterly pro formas are needed for that.

Intuition. EFN is a balancing item, not a forecast of performance. Its sign tells you whether the plan throws off or absorbs cash *at the forecast date*, and anything that adds to assets raises it one-for-one.

Common mistakes.

- Reading a negative EFN as "the forecast is wrong"; it is a cash surplus.
- Believing different forecasting methods give different needs; differences signal an error.
- Using a year-end forecast to manage a seasonal business.

Book answer. 3.1 matches the book (p. 405): "A negative value implies that the company has excess cash above its desired minimum", confirmed by adding it to cash. 3.4: no answer in the book.

Ref: Higgins Ch. 3, "Pro Forma Statements" and chapter summary, pp. 80–102.

3. Corporação Felipe: the plug variable — Ross 4.1 and 4.2, p. 117

What is asked. (4.1) Pro forma statements when sales and *every* balance sheet item grow 15%, and which variable closes the balance sheet. (4.2) Same forecast, but half of net income is paid as dividends and debt and equity do not grow with sales: find the external financing needed (NAF).

Approach. A pro forma model has one degree of freedom too many once we fix sales growth, the percentages, and the financing; one item must be the *variable de fechamento* (plug). In 4.1 the balance sheet is fully specified, so dividends become the plug; in 4.2 the payout is fixed, so the plug moves to external financing.

Notation.

- $g = 15\%$: sales growth; S, C, NI: sales, costs, net income (no taxes).
- A, D, E: assets, debt, equity; Div: dividends; EFN (NAF): external financing needed.

Step 1 — Pro forma income statement (both parts). Costs vary with sales:

$$S_1 = 23{,}000(1.15) = 26{,}450, \qquad C_1 = 16{,}700(1.15) = 19{,}205, \tag{4}$$

$$NI_1 = S_1 - C_1 = 26{,}450 - 19{,}205 = 7{,}245. \tag{5}$$

Step 2 — (4.1) Balance sheet grown 15%. Every item grows 15%: $A_1 = 15{,}800(1.15) = 18{,}170$, $D_1 = 5{,}200(1.15) = 5{,}980$, $E_1 = 10{,}600(1.15) = 12{,}190$, and $5{,}980 + 12{,}190 = 18{,}170$, so it balances.

Step 3 — (4.1) Find the plug. Equity can only change through retained earnings (no share issues), so $E_1 = E_0 + NI_1 - Div_1$. Solving for dividends,

$$Div_1 = NI_1 - (E_1 - E_0) = 7{,}245 - (12{,}190 - 10{,}600) = 7{,}245 - 1{,}590 = 5{,}655. \tag{6}$$

If no dividend were paid, equity would be $10{,}600 + 7{,}245 = 17{,}845$ and the balance sheet would not balance; the dividend of 5,655 is what makes it do so.

Answer (4.1). Sales 26,450, costs 19,205, net income 7,245; assets 18,170 = debt 5,980 + equity 12,190. The plug (*variável de fechamento*) is the dividend, \$5,655.

Step 4 — (4.2) Fixed payout, fixed debt. Dividends are $0.5(7{,}245) = 3{,}622.5$, so retained earnings add 3,622.5 and

$$\begin{aligned} E_1 &= 10{,}600 + 3{,}622.5 = 14{,}222.5, \qquad D_1 = 5{,}200, \\ D_1 + E_1 &= 5{,}200 + 14{,}222.5 = 19{,}422.5, \\ EFN &= A_1 - (D_1 + E_1) = 18{,}170 - 19{,}422.5 = -1{,}252.5. \end{aligned} \tag{7}$$

Answer (4.2). NAF = −\$1,252.5: a surplus. The firm can pay down debt, add to cash, or raise the dividend by that amount.

Intuition. Once growth and the asset intensity are fixed, *something* must give: dividends, borrowing or new equity. Ross' point is that the pro forma does not tell you which; the manager chooses the plug, and the model reports the consequence.

Common mistakes.

- Growing equity by 15% *and* adding net income: equity changes only through retained earnings.
- Reporting a negative NAF as zero; the sign carries information (excess funds).

Ref: Ross §4.3, "O modelo da porcentagem de vendas", pp. 99–104; Exemplo pp. 97–98.

4. Internal and sustainable growth rates — Berk 18.16, p. 561

What is asked. (a) The fastest growth the firm can finance with retained earnings alone; (b) the fastest growth with retained earnings plus the debt that keeps D/E constant; (c) (b) again when 40% of income is paid out.

Approach. Berk's equations (18.4)–(18.5): internal growth = ROA × retention, sustainable growth = ROE × retention, both on *beginning-of-period* assets and equity. With retention b, equity grows by $b \cdot NI$; assets can grow at that rate relative to assets (no external money) or relative to equity (debt grows in step).

Notation.

- $NI = 50{,}000$; $A_0 = 400{,}000$, $E_0 = 250{,}000$: beginning assets and equity.
- $b = 1 -$ payout: retention (plowback) ratio; ROA $= NI/A_0$, ROE $= NI/E_0$.

Step 1 — (a) Internal growth. With payout 0%, $b = 1$:

$$g_{int} = \text{ROA} \times b = \frac{50{,}000}{400{,}000} \times 1 = 0.125 = 12.5\%. \tag{8}$$

Answer (a). 12.5%.

Step 2 — (b) Sustainable growth.

$$g_{sus} = \text{ROE} \times b = \frac{50{,}000}{250{,}000} \times 1 = 0.20 = 20\%. \tag{9}$$

Answer (b). 20%.

Step 3 — (c) Payout 40%. Now $b = 1 - 0.40 = 0.60$:

$$g_{sus} = (0.20)(0.60) = 0.12 = 12\%. \tag{10}$$

Answer (c). 12%.

Intuition. Retained earnings are the only internal source of equity. Measured against assets they bound growth without any outside money; measured against equity they bound growth when debt is allowed to grow in proportion. Leverage ($A_0/E_0 = 1.6$) is why (b) exceeds (a): $20\% = 12.5\% \times 1.6$.

Common mistakes.

- Using end-of-year equity or assets with the simple formulas (then the formulas are $xb/(1-xb)$; see Exercise 13).
- Forgetting to multiply by the retention ratio when there is a payout.

Ref: Berk §18.4, Eqs. (18.4)–(18.5), p. 548; Example 18.3, p. 549.

5. Universal Products: finishing a pro forma — Higgins 3.5, p. 104

What is asked. (a) Net fixed assets at the end of 2022 given \$500 thousand of purchases. (b) Owners' equity with a 50% payout and no equity issues. (c) The resulting external funding required.

Approach. Two roll-forward identities fill the blanks: fixed assets evolve by capex less depreciation; equity by retained earnings. EFN is then the plug of the balance sheet (Exercise 2).

Notation.

- NFA: net fixed assets; $CapEx = 500$; $Dep_{2022} = 260$ (\$ thousands).
- $NI_{2022} = 156$; payout $= 50\%$; E: owners' equity; CA, CL, LTD as in the table.

Step 1 — (a) Fixed assets. Net fixed assets grow by purchases and fall by depreciation:

$$NFA_{2022} = NFA_{2021} + CapEx - Dep_{2022} = 2{,}000 + 500 - 260 = 2{,}240. \tag{11}$$

Answer (a). \$2,240 thousand.

Step 2 — (b) Equity. Equity grows by retained income only:

$$E_{2022} = E_{2021} + NI_{2022}(1 - 0.5) = 2{,}950 + 156(0.5) = 2{,}950 + 78 = 3{,}028. \tag{12}$$

Answer (b). \$3,028 thousand.

Step 3 — (c) The plug. Totals and the difference:

$$\begin{aligned} TA_{2022} &= CA + NFA = 2{,}750 + 2{,}240 = 4{,}990, \\ (L+E)_{2022} &= CL + LTD + E = 1{,}155 + 500 + 3{,}028 = 4{,}683, \\ EFN_{2022} &= 4{,}990 - 4{,}683 = 307. \end{aligned} \tag{13}$$

Answer (c). External funding required = \$307 thousand.

Intuition. Assets grow by $4{,}990 - 4{,}500 = 490$, but spontaneous liabilities supply only $1{,}155 - 1{,}050 = 105$ and retained earnings 78; the remaining $490 - 105 - 78 = 307$ must come from outside.

Common mistakes.

- Adding all of net income to equity (ignoring the 50% dividend).
- Forgetting depreciation in the fixed-asset roll-forward, or subtracting 2021's (250) instead of 2022's (260).

Book answer. Matches the book (p. 405): $NFA = 2{,}240$, equity 3,028, EFN 307.

Ref: Higgins Ch. 3, "Pro Forma Statements", pp. 80–88.

6. Net new financing and how to split it — Berk 18.2 and 18.3, p. 558

What is asked. (18.2) The external financing needed next year given ending assets, net income, payout and the growth of non-debt liabilities. (18.3) With beginning debt of \$100,000, how much of that financing must be debt and how much equity to keep the debt-equity ratio unchanged.

Approach. Berk's identity: Net New Financing = Projected Assets − Projected Liabilities and Equity, where projected liabilities and equity include only the *spontaneous* sources (payables) and retained earnings. Part 18.3 is a financing-policy choice: the total is fixed by 18.2; the policy splits it.

Notation.

- $TA_1 = 500{,}000$; $NI = 50{,}000$; payout 10%; $E_0 = 300{,}000$; $L_0 = 120{,}000$ (of which $D_0 = 100{,}000$ debt in 18.3).
- NNF: net new financing; ΔD, ΔE^{new}: new debt and new equity issued.

Step 1 — (18.2) Projected sources. Retained earnings are $50{,}000(1 - 0.10) = 45{,}000$, so

$$E_1 = 300{,}000 + 45{,}000 = 345{,}000, \qquad L_1 = 120{,}000 + 10{,}000 = 130{,}000,$$
$$NNF = TA_1 - (L_1 + E_1) = 500{,}000 - (130{,}000 + 345{,}000) = 500{,}000 - 475{,}000 = 25{,}000. \quad (14)$$

Answer (18.2). Net new financing = \$25,000.

Step 2 — (18.3) Target capital structure. Non-debt liabilities are $120{,}000 - 100{,}000 = 20{,}000$ at the start and $20{,}000 + 10{,}000 = 30{,}000$ at the end. Debt plus equity must therefore fund $500{,}000 - 30{,}000 = 470{,}000$ of assets, in the initial proportion $D/E = 100{,}000/300{,}000 = 1/3$, i.e. $D = \frac{1}{4}$ and $E = \frac{3}{4}$ of the total:

$$E_1 = \tfrac{3}{4}(470{,}000) = 352{,}500, \qquad D_1 = \tfrac{1}{4}(470{,}000) = 117{,}500. \quad (15)$$

Step 3 — (18.3) Issues. Equity already reaches 345,000 through retention; debt starts at 100,000:

$$\Delta E^{new} = 352{,}500 - 345{,}000 = 7{,}500, \qquad \Delta D = 117{,}500 - 100{,}000 = 17{,}500,$$

and $7{,}500 + 17{,}500 = 25{,}000 = NNF$.

Answer (18.3). Issue \$7,500 of equity and \$17,500 of debt.

Intuition. Net new financing is fixed by the operating plan; how it is raised is a separate capital structure decision. Keeping D/E constant forces some equity issuance here because retained earnings (45,000) are too small relative to the asset growth.

Common mistakes.

- Treating payables as debt when computing D/E; they are spontaneous operating liabilities.
- Applying $D/E = 1/3$ to the \$25,000 alone; the ratio must hold for the *stocks*, and retained earnings already added equity.

Ref: Berk §18.2, "Pro Forma Balance Sheet" and "Making the Balance Sheet Balance", pp. 539–541.

7. Companhia João Herdeiro: percent-of-sales balance sheet — Ross 4.9–4.11, pp. 118–119

What is asked. (4.9) The pro forma income statement at 20% sales growth with constant payout, and the addition to retained earnings. (4.10) Each balance sheet item as a percent of sales, "n/a"

where it does not vary with sales. (4.11) The pro forma balance sheet at 15% growth with no new debt or equity, and the NAF.

Approach. Percent of sales: costs, assets and payables scale with sales; notes payable, long-term debt and paid-in capital stay; retained earnings grow by the retained part of the new net income. NAF is the plug.

Notation.

- $S_0 = 38{,}000$, $C_0 = 18{,}400$; tax rate 34%; $Div_0 = 5{,}200$, $NI_0 = 12{,}936$.
- $b = 7{,}736/12{,}936 = 0.598021$: retention ratio; $1 - b = 0.401979$: payout.

Step 1 — (4.9) Income statement at $g = 20\%$.

$$S_1 = 38{,}000(1.2) = 45{,}600, \quad C_1 = 18{,}400(1.2) = 22{,}080, \quad EBT_1 = 45{,}600 - 22{,}080 = 23{,}520,$$
$$T_1 = 0.34(23{,}520) = 7{,}996.8, \qquad NI_1 = 23{,}520 - 7{,}996.8 = 15{,}523.2. \tag{16}$$

Since all lines scale with sales, $NI_1 = 1.2(12{,}936) = 15{,}523.2$ too. With a constant payout, $Div_1 = 15{,}523.2(0.401979) = 6{,}240$ and the addition to retained earnings is $15{,}523.2(0.598021) = 9{,}283.2$ $(= 1.2 \times 7{,}736)$.

Answer (4.9). Addition to retained earnings = \$9,283.20 (dividends \$6,240).

Step 2 — (4.10) Percent of sales. Divide each item by $S_0 = 38{,}000$:

Ativo	\$	% sales	Passivo e PL	\$	% sales
Caixa	3,050	8.03	Contas a pagar	1,300	3.42
Contas a receber	6,900	18.16	Empréstimos	6,800	n/a
Estoque	7,600	20.00	Total circulante	8,100	n/a
Total circulante	17,550	46.18	Passivo não circulante	25,000	n/a
Instalações e equip.	34,500	90.79	Capital social	15,000	n/a
			Lucros retidos	3,950	n/a
			Total PL	18,950	n/a
Total do ativo	52,050	136.97	Total	52,050	n/a

The current-liabilities total is n/a because only part of it (payables) varies with sales.

Answer (4.10). See the table: all assets and payables scale (total assets 136.97% of sales); loans, long-term debt and equity are n/a.

Step 3 — (4.11) Income at $g = 15\%$. $S_1 = 43{,}700$, $C_1 = 21{,}160$, $EBT_1 = 22{,}540$, $T_1 = 0.34(22{,}540) = 7{,}663.6$, $NI_1 = 22{,}540 - 7{,}663.6 = 14{,}876.4$; dividends $14{,}876.4(0.401979) = 5{,}980$; addition to retained earnings $14{,}876.4(0.598021) = 8{,}896.4$.

Step 4 — (4.11) Pro forma balance sheet. Scaling assets and payables by 1.15 and rolling retained earnings forward $(3{,}950 + 8{,}896.4 = 12{,}846.4)$:

Ativo	\$	Passivo e PL	\$
Caixa	3,507.5	Contas a pagar	1,495.0
Contas a receber	7,935.0	Empréstimos	6,800.0
Estoque	8,740.0	Total circulante	8,295.0
Total circulante	20,182.5	Passivo não circulante	25,000.0
Instalações e equip.	39,675.0	Capital social	15,000.0
		Lucros retidos	12,846.4
		Total PL	27,846.4
Total do ativo	59,857.5	Total	61,141.4

$$\text{NAF} = 59{,}857.5 - (8{,}295 + 25{,}000 + 27{,}846.4) = 59{,}857.5 - 61{,}141.4 = -1{,}283.9. \tag{17}$$

Answer (4.11). NAF = −\$1,283.90: at 15% growth the firm generates a surplus.

Intuition. Asset needs rise by $0.15 \times 52{,}050 = 7{,}807.5$; payables supply 195 and retained earnings 8,896.4, more than enough. A very profitable firm (net margin 34%) can finance moderate growth internally; the surplus would turn into a deficit at a higher growth rate (Exercise 14 shows how to find the switch point).

Common mistakes.

- Using the 20% income statement of 4.9 in 4.11; the growth rate changes to 15%.
- Scaling loans (*empréstimos*) with sales: they are a financing decision.

Ref: Ross §4.3, pp. 99–104.

8. Global Corp.: two-pass pro forma — Berk 18.8–18.11, p. 560

What is asked. (18.8) Next year's costs, depreciation, net income and the operating balance sheet items at 8% sales growth. (18.9) Equity with a 50% payout. (18.10) Net new financing. (18.11) The payout that keeps net new financing at or below \$9 million.

Approach. Berk's KMS template (Tables 18.2–18.3): costs and depreciation as a percent of sales, interest expense unchanged, tax at the current rate; assets and payables as a percent of sales; debt unchanged in the first pass; equity rolls forward by retained earnings; net new financing is the gap. Part 18.11 inverts the plug: fix NNF, solve for retention.

Notation.

- $g = 8\%$; amounts in \$ millions; $\tau = 0.7/2.7 = 0.259259$: Global's tax rate (taxes/pretax income).
- p: payout; NNF: net new financing.

Step 1 — (18.8) Income statement. Every operating line scales by 1.08; interest stays at 7.7:

$$\begin{aligned} S_1 &= 186.7(1.08) = 201.636, \quad C_1 = 175.1(1.08) = 189.108, \quad Dep_1 = 1.2(1.08) = 1.296, \\ EBIT_1 &= 201.636 - 189.108 - 1.296 = 11.232, \qquad EBT_1 = 11.232 - 7.7 = 3.532, \\ T_1 &= 0.259259(3.532) = 0.915704, \qquad NI_1 = 3.532 - 0.915704 = 2.616296. \end{aligned} \tag{18}$$

(Depreciation is $1.2/186.7 = 0.6427\%$ of sales; scaling by 1.08 is the same thing.)

Step 2 — (18.8) Balance sheet items. Each scales by 1.08: cash $23.2(1.08) = 25.056$; receivables $18.5(1.08) = 19.98$; inventory $15.3(1.08) = 16.524$ (current assets 61.56); PP&E $113.1(1.08) = 122.148$; total assets 183.708; payables $34.7(1.08) = 37.476$.

Answer (18.8). (a) 189.108; (b) 1.296; (c) 2.616; (d) 25.056; (e) 19.98; (f) 16.524; (g) 122.148; (h) 37.476 (\$ millions).

Step 3 — (18.9) Equity. Retained earnings are $0.5(2.616296) = 1.308148$:

$$E_1 = 22.2 + 1.308148 = 23.508148. \tag{19}$$

Answer (18.9). Stockholders' equity ≈ \$23.51 million.

Step 4 — (18.10) Net new financing. Debt stays at 113.2 in the first pass:

$$\begin{aligned} (L+E)_1 &= 37.476 + 113.2 + 23.508148 = 174.184148, \\ NNF &= 183.708 - 174.184148 = 9.523852. \end{aligned} \tag{20}$$

Answer (18.10). Net new financing ≈ \$9.52 million.

Step 5 — (18.11) Payout cap. Write NNF as a function of the payout: $NNF(p) = 183.708 - 37.476 - 113.2 - 22.2 - (1-p)NI_1 = 10.832 - (1-p)(2.616296)$. Requiring $NNF \le 9$:

$$\begin{aligned}(1-p)(2.616296) &\ge 10.832 - 9 = 1.832,\\ 1-p &\ge \frac{1.832}{2.616296} = 0.700227,\\ p &\le 0.299773.\end{aligned} \tag{21}$$

Dividends fall from 1.308 to at most $2.616296 - 1.832 = 0.784$ million, i.e. by the 0.524 excess.

Answer (18.11). Global must cut its payout from 50% to at most about 30% (dividends $\le$ \$0.78 million).

Intuition. Global is highly levered with a thin margin (net income \$2.0 million on \$186.7 million of sales), so 8% growth in a \$170 million asset base creates a financing need (13.6 of new assets) that retained earnings barely dent. The payout is the only internal lever left; cutting it from 50% to 30% saves just \$0.52 million.

Common mistakes.

- Growing interest expense with sales; the method holds it fixed until the financing is decided.
- Using a 35% tax rate (the KMS rate) instead of Global's own 0.7/2.7.
- Solving 18.11 by scaling dividends instead of solving the inequality for the payout.

Ref: Berk §18.2, Tables 18.1–18.4 and Example 18.2, pp. 537–541.

9. KMS growth vs. its SGR; new debt; D/E at the internal rate — Berk 18.17–18.19, p. 561

What is asked. (18.17) Whether the KMS plan grows sales faster or slower than its sustainable growth rate (SGR). (18.18) How much debt a firm must issue if it grows exactly at its SGR. (18.19) IZAX's new D/E if its assets grow at its internal growth rate.

Approach. SGR = ROE $\times b$ (beginning equity), Eq. (18.5): the growth that leaves D/E constant with no new equity. Comparing planned growth with SGR tells us whether leverage, payout or equity issues must change. At the internal rate no external money is raised, so debt is frozen and D/E falls.

Notation.

- $g_t = S_t/S_{t-1} - 1$: KMS sales growth; $\text{SGR}_t = (NI_t - Div_t)/E_{t-1}$; $\text{ROE}_t = NI_t/E_{t-1}$.
- 18.18: ROE = 12%, payout 25%, $E_0 = 600{,}000$, $D_0 = 400{,}000$.
- 18.19: $A_0 = 400{,}000$, $D_0 = 100{,}000$, $E_0 = 300{,}000$, $NI = 20{,}000$, $Div = 5{,}000$.

Step 1 — (18.17) Growth and SGR year by year. Using Tables 18.8 and 18.17 (\$000s); e.g. 2011: $g = 88{,}369/74{,}889 - 1 = 18.00\%$ and $\text{SGR} = (7{,}600 - 4{,}786)/74{,}134 = 2{,}814/74{,}134 = 3.80\%$.

	2011	2012	2013	2014	2015
Sales growth g_t (%)	18.00	16.84	16.03	15.34	14.75
SGR with the plan's dividends (%)	3.80	8.19	8.29	8.40	8.51
ROE = SGR at full retention (%)	10.25	11.45	13.38	15.24	17.01

Even under the 2010 policy (30% payout, Section 18.2) the SGR was $6{,}941/69{,}275 \times 0.7 = 7.01\%$.

Answer (18.17). Faster: sales grow 15–18% a year against an SGR of 4–9%. KMS covers the gap by raising leverage — book D/E jumps from $4{,}500/74{,}134 = 6.07\%$ to $24{,}500/76{,}948 = 31.84\%$ in 2011 — and, through 2014, growth exceeds even the full-retention ROE.

Step 2 — (18.18) Debt at the SGR.

$$\begin{aligned} &\text{SGR} = 0.12(1-0.25) = 0.09, \qquad \Delta E = 600{,}000(0.12)(0.75) = 54{,}000 = 9\% \text{ of } E_0, \\ &\Delta D = 0.09(400{,}000) = 36{,}000. \end{aligned} \tag{22}$$

Check: $(400{,}000 + 36{,}000)/(600{,}000 + 54{,}000) = 436{,}000/654{,}000 = 2/3 = 400{,}000/600{,}000$.

Answer (18.18). Issue \$36,000 of new debt.

Step 3 — (18.19) IZAX at the internal rate.

$$\begin{aligned} &\text{ROA} = 20{,}000/400{,}000 = 0.05, \quad b = 15{,}000/20{,}000 = 0.75, \quad g_{int} = (0.05)(0.75) = 0.0375, \\ &A_1 = 400{,}000(1.0375) = 415{,}000, \quad E_1 = 300{,}000 + 15{,}000 = 315{,}000, \quad D_1 = 100{,}000, \\ &D_1/E_1 = 100{,}000/315{,}000 = 0.3175. \end{aligned} \tag{23}$$

Answer (18.19). $D/E = 0.3175$ (down from 0.3333).

Intuition. The SGR is a benchmark, not a target. KMS's planned growth is far above it, which only says that the plan needs outside money (here a \$20 million bond); whether that growth is *worth* it is an NPV question (Exercise 15). At the internal rate equity grows and debt does not, so leverage drifts down; at the sustainable rate debt grows with equity and leverage stays put.

Common mistakes.

- Calling growth above the SGR "bad": Berk stresses that only NPV decides value.
- Computing ROE on end-of-year equity with the simple formula.

Ref: Berk §18.4, pp. 547–551; Tables 18.8, 18.17, pp. 545, 562.

10. Under Armour: sustainable growth and its sensitivity — Higgins 4.8, pp. 135–136

What is asked. (a) The sustainable growth rate each year 2009–2013. (b) Whether actual growth outran it. (c) How the company coped. (d) The 2013 SGR with an 8% margin, with leverage of 2.40, and with both.

Approach. Higgins' $g^* = PRA\hat{T}$: profit margin × retention × asset turnover × assets/beginning equity. A firm growing faster than g^* must improve operations (P, A) or change financial policy (R, $\hat{T}$) or issue equity. Part (d) is a one-at-a-time sensitivity analysis.

Notation.

- P (%), $R = 1$, A (times), $\hat{T}$ (times): the four ratios in the table; g: actual sales growth.

Step 1 — (a) The product of four ratios. With $R = 1$ in every year, $g^* = P \cdot A \cdot \hat{T}$:

$$\begin{aligned} &g^*_{2009} = 5.5(1.6)(1.7) = 14.96\%, \qquad &&g^*_{2010} = 6.4(1.6)(1.7) = 17.41\%, \\ &g^*_{2011} = 6.6(1.6)(1.9) = 20.06\%, \qquad &&g^*_{2012} = 7.0(1.6)(1.8) = 20.16\%, \\ &g^*_{2013} = 7.0(1.5)(1.9) = 19.95\%. \end{aligned} \tag{24}$$

Answer (a). 14.96%, 17.41%, 20.06%, 20.16%, 19.95%.

Step 2 — (b) Compare with actual growth. Actual growth was 18.1, 24.2, 38.4, 24.6 and 27.1%: above g^* every year, by up to $38.4 - 20.06 = 18.34$ points in 2011.

Answer (b). Yes: a too-much-growth challenge — growing faster than equity could be built internally.

Step 3 — (c) Which ratios moved. Retention was already 100%. The margin rose (5.5% → 7.0%) and leverage rose ($\hat{T}$: 1.7 → 1.9); turnover slipped (1.6 → 1.5), which worked the other way.

Because each year's g^* uses that year's ratios, growth above g^* in a year must have been met by ratios rising further (or new equity) during that year.

Answer (c). By raising profitability and, above all, financial leverage (more debt per dollar of beginning equity), with no dividends; operating efficiency (turnover) did not help.

Step 4 — (d) Sensitivity of 2013 g^*.

$$\begin{aligned} P = 8.0\% : &\quad g^* = 8.0(1.5)(1.9) = 22.8\%, \\ \hat{T} = 2.40 : &\quad g^* = 7.0(1.5)(2.4) = 25.2\%, \\ \text{both} : &\quad g^* = 8.0(1.5)(2.4) = 28.8\%. \end{aligned} \tag{25}$$

Answer (d). 22.8%, 25.2% and 28.8%; only the combination exceeds 2013's actual 27.1%.

Intuition. g^* is linear in each lever, so the levers multiply: a 14% rise in the margin and a 26% rise in leverage together lift g^* by 44%. Leverage is the easiest lever to pull and the riskiest to rely on.

Common mistakes.

- Adding the percentage effects in (d) instead of multiplying the ratios.
- Using end-of-period assets/equity; Higgins' $\hat{T}$ uses beginning equity.

Ref: Higgins Ch. 4, "Sustainable Growth" and "What If Actual Growth Exceeds Sustainable Growth?", pp. 112–123.

11. Turismo da Onça: NAF, capacity, constant D/E — Ross 4.25–4.27, p. 121

What is asked. (4.25) The NAF for 20% growth at full capacity with no new debt or equity, interest, tax rate and payout constant. (4.26) The NAF if the firm ran at 80% of capacity. (4.27) The NAF if the firm keeps its debt-equity ratio constant.

Approach. Percent of sales for costs, other expenses, current assets and payables; fixed assets scale only if there is no idle capacity; interest is *fixed*. In 4.27 the financing policy allows debt to grow with equity, so part of the gap is filled by planned borrowing.

Notation.

- $S_0 = 929{,}000$; $\tau = 60{,}550/173{,}000 = 0.35$; payout $= 33{,}735/112{,}450 = 0.30$; interest 14,000.
- "Total debt" (Ross) = all liabilities: $D_0 = 85{,}000 + 158{,}000 = 243{,}000$; $E_0 = 322{,}900$.

Step 1 — Income statement at $g = 20\%$.

$$\begin{gathered} S_1 = 1{,}114{,}800, \quad \text{Custos} = 867{,}600, \quad \text{Outras} = 22{,}800, \quad EBIT = 224{,}400, \\ EBT = 224{,}400 - 14{,}000 = 210{,}400, \quad T = 0.35(210{,}400) = 73{,}640, \quad NI = 136{,}760, \\ Div = 0.30(136{,}760) = 41{,}028, \qquad \text{Acréscimo aos lucros retidos} = 95{,}732. \end{gathered} \tag{26}$$

Step 2 — (4.25) Balance sheet at full capacity. Current assets $152{,}900(1.2) = 183{,}480$ (cash 30,360, receivables 48,840, inventory 104,280); fixed assets $413{,}000(1.2) = 495{,}600$; total 679,080. Payables 81,600; notes 17,000; long-term 158,000; equity $140{,}000 + 182{,}900 + 95{,}732 = 418{,}632$; total $81{,}600 + 17{,}000 + 158{,}000 + 418{,}632 = 675{,}232$.

$$\text{NAF} = 679{,}080 - 675{,}232 = 3{,}848. \tag{27}$$

Answer (4.25). NAF = \$3,848.

Step 3 — (4.26) Idle capacity. Full-capacity sales are 929,000/0.80 = 1,161,250 > 1,114,800, so no new fixed assets are needed (utilization rises to 96%). Total assets = 183,480 + 413,000 = 596,480:

$$\text{NAF} = 596{,}480 - 675{,}232 = -78{,}752. \tag{28}$$

Answer (4.26). NAF = −$78,752 (a surplus).

Step 4 — (4.27) Constant total-debt/equity. $D/E = 243{,}000/322{,}900 = 0.752555$. With equity 418,632, permitted total debt is 0.752555(418,632) = 315,043.59; total sources 418,632 + 315,043.59 = 733,675.59:

$$\text{NAF} = 679{,}080 - 733{,}675.59 = -54{,}595.59. \tag{29}$$

Existing and spontaneous liabilities total 81,600 + 17,000 + 158,000 = 256,600, so the policy allows 315,043.59 − 256,600 = 58,443.59 of new borrowing, which more than covers the 3,848 gap.

Answer (4.27). NAF ≈ −$54,596: the debt the policy allows exceeds the need by that amount.

Intuition. Three different answers from one forecast: the NAF depends as much on assumptions about capacity and financing policy as on the sales forecast. Idle capacity is worth $82,600 of financing here (495,600 − 413,000).

Common mistakes.

- Growing interest expense with sales (the statement holds it constant).
- In 4.26, cutting fixed assets below the current level; they stay at 413,000.
- In 4.27, applying D/E only to interest-bearing debt; Ross' ratio uses total liabilities (the alternative definition changes the number, so state it).

Ref: Ross §4.3, "Uma situação alternativa", pp. 103–104; §4.4, pp. 106–111.

12. R&E Supplies: extending the spreadsheet, sensitivity and scenario — Higgins 3.9 (3.8), p. 105

What is asked. (3.8, Part 0) The 2023 spreadsheet formulas for net sales and equity. (a) The 2023 external financing required (EFR) and its comparison with 2022. (b) EFR if COGS falls to 84% of sales. (c) EFR in a recession: sales −5%, COGS 88%, current assets 35% of sales.

Approach. Extend Table 3.5 one column: 2023 sales grow from 2022 forecast sales; 2023 equity starts from 2022 forecast equity. Interest is charged on long-term debt, its current portion *and* the EFR itself, which makes EFR circular; we solve the circularity in closed form instead of iterating. (b) changes one input (sensitivity); (c) changes several together (scenario).

Notation.

- X: 2023 EFR ($ thousands); $i = 10\%$; $\tau = 25\%$; payout 50%; $LTD = 560$, $CPLTD = 100$, $NFA = 270$.
- S, GP, SGA, CA, CL: 2023 sales, gross profit, SG&A (11%), current assets, current liabilities (14.4%).
- $S_{22} = 20{,}613(1.25) = 25{,}766.25$ and $E_{22} = 1{,}837.81$: Table 3.5's unrounded 2022 values (the table prints 25,766 and 1,838).

Step 0 — (3.8) Formulas. `D19 = C19 + C19*D4` (2023 sales from 2022 forecast sales) and `D37 = C37 + D28` (2023 equity = 2022 forecast equity + 2023 additions to retained earnings).

Answer (3.8). `D19 = C19 + C19*D4`; `D37 = C37 + D28`.

Step 1 — Closed form for the circular interest. EFR $= TA - CL - LTD - E$, with $E = E_{22} + (1 - 0.5)(1 - \tau)[GP - SGA - i(LTD + CPLTD + X)]$. Let $k = (0.5)(0.75) = 0.375$. Then

$$\begin{aligned} X &= TA - CL - LTD - E_{22} - k[GP - SGA - i(660)] + k\,i\,X, \\ X(1 - k\,i) &= \underbrace{TA - CL - LTD - E_{22} - k[GP - SGA - 66]}_{K}, \qquad X = \frac{K}{1 - 0.0375} = \frac{K}{0.9625}. \end{aligned} \tag{30}$$

Step 2 — (a) Base case. $S = 25{,}766.25(1.30) = 33{,}496.13$; $COGS = 0.86S = 28{,}806.67$; $GP = 4{,}689.46$; $SGA = 0.11S = 3{,}684.57$; $CA = 0.29S = 9{,}713.88$; $TA = 9{,}713.88 + 270 = 9{,}983.88$; $CL = 0.144S = 4{,}823.44$.

$$\begin{aligned} K &= 9{,}983.88 - 4{,}823.44 - 560 - 1{,}837.81 - 0.375(4{,}689.46 - 3{,}684.57 - 66) = 2{,}410.54, \\ X &= 2{,}410.54/0.9625 = 2{,}504.46. \end{aligned} \tag{31}$$

Then interest $= 0.1(660 + 2{,}504.46) = 316.45$; EBT $= 4{,}689.46 - 3{,}684.57 - 316.45 = 688.44$; tax 172.11; EAT 516.33; dividends 258.16; equity $1{,}837.81 + 258.16 = 2{,}095.97$; liabilities and equity $4{,}823.44 + 560 + 2{,}095.97 = 7{,}479.42$; and $9{,}983.88 - 7{,}479.42 = 2{,}504.46 = X$, as it must.

Answer (a). EFR ≈ \$2,504 thousand in 2023, about \$986 thousand more than 2022's \$1,518 thousand.

Step 3 — (b) Sensitivity: COGS 84%. Only GP changes: $GP = 0.16S = 5{,}359.38$, so $K = 2{,}410.54 - 0.375(5{,}359.38 - 4{,}689.46) = 2{,}159.32$ and $X = 2{,}159.32/0.9625 = 2{,}243.45$.

Answer (b). EFR ≈ \$2,243 thousand: −\$261 thousand, or −10.4%.

Step 4 — (c) Recession scenario. $S = 25{,}766.25(0.95) = 24{,}477.94$; $COGS = 0.88S = 21{,}540.59$; $GP = 2{,}937.35$; $SGA = 2{,}692.57$; $CA = 0.35S = 8{,}567.28$; $TA = 8{,}837.28$; $CL = 3{,}524.82$.

$$\begin{aligned} K &= 8{,}837.28 - 3{,}524.82 - 560 - 1{,}837.81 - 0.375(2{,}937.35 - 2{,}692.57 - 66) = 2{,}847.60, \\ X &= 2{,}847.60/0.9625 = 2{,}958.55. \end{aligned} \tag{32}$$

Interest = 361.85, EBT = −117.08, EAT = −87.81, and the spreadsheet's dividend formula `=C11*C26` pays −43.90 (a negative dividend).

Answer (c). EFR ≈ \$2,959 thousand, +18.1% over the base case. If dividends are set to zero when earnings are negative (the realistic policy), EFR is ≈ \$3,006 thousand.

Intuition. The financing need is far more sensitive to working capital than to margins: the recession *lowers* sales but raises EFR because current assets jump to 35% of a smaller sales figure and profits vanish. Sensitivity analysis shows which inputs drive the answer; scenario analysis shows how they move together.

Common mistakes.

- Starting 2023 from 2021 data (sales \$20,613, equity \$1,730) instead of the 2022 forecast.
- Ignoring the interest on the EFR itself (understates EFR by about $0.0375X \approx$ \$94 thousand).
- Copying a formula that pays negative dividends without noticing.

Book answer. Matches the book (p. 406): \$2.504 million; \$2.243 million (−10.4%); \$2.959 million (+18.2% in the book; +18.1% here from unrounded values). The book's recession figure uses the spreadsheet's mechanical (negative) dividend.

Ref: Higgins Ch. 3, "Forecasting with Spreadsheets" and "Coping with Uncertainty", pp. 89–98.

13. Deriving NAF and the growth formulas — Ross 4.31–4.33, p. 122

What is asked. (4.31) Show NAF $= -ML(V)b + (A - ML(V)b)g$. (4.32) Use it to recover the internal and sustainable growth formulas [4.2]–[4.3]. (4.33) Show that the end-of-period formula $\text{ROE}\,b/(1-\text{ROE}\,b)$ equals $\text{ROE}_{bop} \times b$, and similarly for ROA.

Approach. The NAF is uses minus internal sources. All assets scale with sales (asset need Ag), the margin is constant, and no liability is spontaneous; retained earnings are next year's income times b. The sustainable case adds the debt that keeps D/P constant.

Notation.

- V: last year's sales; A, D, P: last year's total assets, debt, equity ($A = D + P$).
- g: sales growth; ML: profit margin; b: retention; $\text{ROA} = ML \cdot V/A$, $\text{ROE} = ML \cdot V/P$ (Ross' definitions, end of the base year).

Step 1 — (4.31) Uses and sources. New assets are Ag; next year's net income is $ML \cdot V(1+g)$, of which b is retained. So

$$\begin{aligned} \text{NAF} &= Ag - ML(V)b(1+g) \\ &= Ag - ML(V)b - ML(V)b\,g \\ &= -ML(V)b + (A - ML(V)b)g. \end{aligned} \tag{33}$$

Answer (4.31). Shown in (33).

Step 2 — (4.32) Internal growth. Set NAF $= 0$ in (33):

$$(A - ML(V)b)g = ML(V)b \implies g = \frac{ML(V)b}{A - ML(V)b}.$$

Divide numerator and denominator by A and use $\text{ROA} = ML \cdot V/A$:

$$g_{int} = \frac{(ML \cdot V/A)\,b}{1 - (ML \cdot V/A)\,b} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}. \tag{34}$$

Step 3 — (4.32) Sustainable growth. Keeping D/P constant lets debt rise by (D/P) per dollar of retained earnings, so the sources are $(1 + D/P)\,ML(V)b(1+g)$ and $1 + D/P = A/P$. Setting uses equal to sources:

$$\begin{aligned} Ag = \frac{A}{P}ML(V)b(1+g) &\implies g\Big(A - \frac{A}{P}ML(V)b\Big) = \frac{A}{P}ML(V)b, \\ g_{sus} = \frac{(ML \cdot V/P)\,b}{1 - (ML \cdot V/P)\,b} &= \frac{\text{ROE} \times b}{1 - \text{ROE} \times b} \quad \text{(dividing by } A\text{)}. \end{aligned} \tag{35}$$

Illustration (Ross' Hoffman example: $V = A = 500$, $NI = 66$, $P = 250$, $b = 2/3$): $g_{int} = 9.65\%$, $g_{sus} = 21.36\%$.

Answer (4.32). Both chapter formulas [4.2] and [4.3] follow from (33).

Step 4 — (4.33) Beginning- vs. end-of-period ROE. With no new shares, end equity is $P_e = P_b + b\,NI$. Let $\text{ROE}_e = NI/P_e$. Then

$$\frac{\text{ROE}_e\,b}{1 - \text{ROE}_e\,b} = \frac{b\,NI/P_e}{1 - b\,NI/P_e} = \frac{b\,NI}{P_e - b\,NI} = \frac{b\,NI}{P_b} = \text{ROE}_b \times b. \tag{36}$$

The same steps with assets work when assets grow only by retained earnings (no external financing, which is exactly the internal-growth case), $A_e = A_b + b\,NI$: $\text{ROA}_e b/(1-\text{ROA}_e b) = b\,NI/(A_e - b\,NI) = b\,NI/A_b = \text{ROA}_b \times b$. Ross' numbers: $NI = 20$, $b = 0.6$, $P_b = 80$ give $0.6(20/80) = 15\%$ and $0.6(0.2174)/(1 - 0.6(0.2174)) = 15\%$; with $A_b = 100$, 12%.

Answer (4.33). (36) and its ROA analogue.

Intuition. The "$1 - \text{ROE}\, b$" denominator is just the conversion from end-of-period to beginning-of-period equity: retained earnings are in the end balance but cannot have financed themselves. Use one consistent definition and you get the same growth rate.

Common mistakes.

- Forgetting the $(1+g)$ on next year's earnings, which drops the $-ML(V)b\,g$ term.
- Mixing definitions: end-of-period ROE in the simple ROE$\times b$ understates growth ($0.6\times 0.2174 = 13.0\%$ instead of 15% in Ross' example); beginning-of-period ROE in [4.3] overstates it ($0.15/(1-0.15) = 17.6\%$).

Ref: Ross §4.4, pp. 106–113, Tabela 4.9.

14. Turismo da Onça: NAF as a function of growth — Ross 4.28 and 4.29, p. 121

What is asked. (4.28) Redo 4.25 at 15% and 25% growth, plot NAF against g, find the g where NAF $= 0$, and explain why it differs from the internal growth formula. (4.29) The same for 4.27 at 20, 30 and 35%.

Approach. Write NAF as an explicit function of g using the model of Exercise 11: assets and payables scale with $1+g$; interest is fixed at 14,000; retained earnings $= (1-0.30)(1-0.35)\big[187{,}000(1+g) - 14{,}000\big] = 0.455\big[187{,}000(1+g) - 14{,}000\big]$. NAF is then linear in g, and its root is the model's own internal (4.28) or sustainable (4.29) rate.

Notation.

- $RE(g)$: addition to retained earnings at growth g; $D/E = 0.752555$ (total liabilities/equity).

Step 1 — (4.28) NAF(g) with no new debt or equity.

$$\begin{aligned}\text{NAF}(g) &= 565{,}900(1+g) - \big[68{,}000(1+g) + 17{,}000 + 158{,}000 + 322{,}900 + RE(g)\big] \\ &= (565{,}900 - 68{,}000 - 0.455\cdot 187{,}000)(1+g) - (497{,}900 - 0.455\cdot 14{,}000) \\ &= 412{,}815(1+g) - 491{,}530. \end{aligned} \tag{37}$$

At $g = 15\%$: $RE = 91{,}477.75$, NAF $= -16{,}792.75$; at 20%: $RE = 95{,}732$, NAF $= 3{,}848$; at 25%: $RE = 99{,}986.25$, NAF $= 24{,}488.75$. Root: $1+g = 491{,}530/412{,}815$, so $g = 19.07\%$.

Step 2 — (4.28) The textbook formula. ROA $= 112{,}450/565{,}900 = 19.87\%$, ROA$\cdot b = 0.139097$, $g_{int} = 0.139097/(1-0.139097) = 16.16\%$. The formula equals the root of $565{,}900g - 78{,}715(1+g) = 0$, i.e. it assumes (i) no spontaneous liabilities and (ii) a constant net margin. Adding (i) back (payables grow with sales) moves the root to $78{,}715/(565{,}900 - 68{,}000 - 78{,}715) = 18.78\%$; adding (ii) (fixed interest makes the margin rise with sales) moves it to 19.07%.

Answer (4.28). NAF $= -16{,}793$; 3,848; 24,489. NAF $= 0$ at $g \approx 19.07\%$, above the formula's 16.16% because payables finance part of the growth and fixed interest raises the margin.

Step 3 — (4.29) NAF(g) with constant D/E. Sources are equity times $(1 + D/E)$:

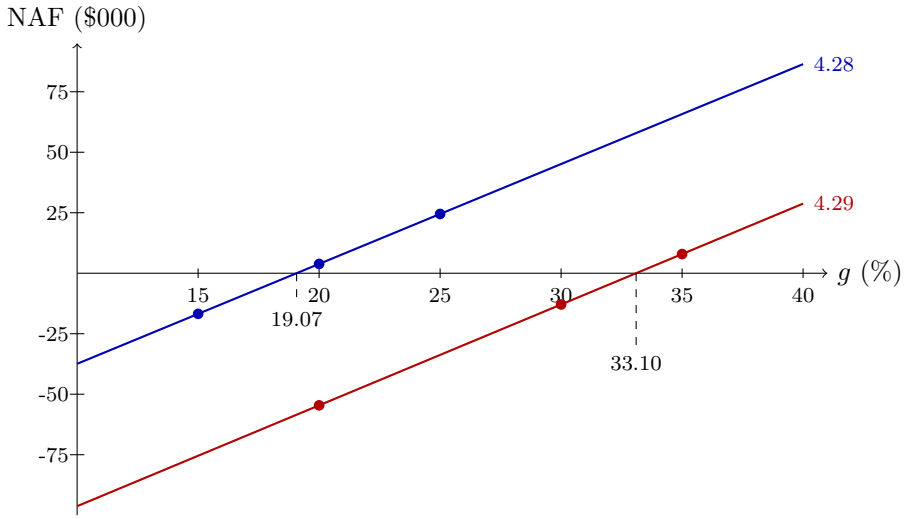
$$\begin{aligned}\text{NAF}(g) &= 565{,}900(1+g) - 1.752555\big[322{,}900 + RE(g)\big] \\ &= (565{,}900 - 1.752555\cdot 85{,}085)(1+g) - 1.752555(322{,}900 - 6{,}370) \\ &= 416{,}783.86(1+g) - 554{,}736.22. \end{aligned} \tag{38}$$

At 20%: NAF $= -54{,}595.59$; at 30% ($RE = 104{,}240.5$): $-12{,}917.21$; at 35% ($RE = 108{,}494.75$): 7,921.99. Root: $g = 554{,}736.22/416{,}783.86 - 1 = 33.10\%$.

Step 4 — (4.29) The textbook formula. ROE $= 112{,}450/322{,}900 = 34.83\%$, ROE $\cdot\, b = 0.243775$, $g_{sus} = 0.243775/(1-0.243775) = 32.24\%$, the root of $565{,}900g = 1.752555 \times 78{,}715(1+g)$

$= 137{,}952.36(1+g)$, which assumes a constant margin. Payables do not matter here (they count as debt in D/E); the gap is the fixed interest only.

Answer (4.29). NAF $= -54{,}596$; $-12{,}917$; $7{,}922$. NAF $= 0$ at $g \approx 33.10\%$, above the formula's 32.24% because interest does not grow with sales, so the margin rises with growth.



Intuition. NAF rises linearly with growth: each extra point of growth needs \$4,128 more financing in 4.28 (412,815/100) and \$4,168 in 4.29. The closed-form growth rates are exact only in the stylized world where every cost scales with sales; real pro formas have fixed costs (here interest), so always verify with the model itself.

Common mistakes.

- Treating the formula rates as the model's break-even rates without checking its assumptions.
- Plotting NAF against g with curvature: in this model it is a straight line.

Ref: Ross §4.4, "Necessidade de aporte financeiro e crescimento", pp. 106–111.

15. KMS Designs under 0.25% share growth — Berk 18.20–18.22 (18.12–18.15), pp. 560–561

What is asked. (18.12) Required production each year and when it first exceeds the 1,100 thousand-unit capacity. (18.13) Interest and interest tax shields through 2015 when the \$20 million expansion is delayed to 2012. (18.14) Net income through 2015 with the given depreciation. (18.15) Working capital needs. (18.20) Free cash flows. (18.21) Continuation value at 8.5 times 2015 EBITDA. (18.22) Firm value at a 10% cost of capital.

Approach. Rebuild the KMS model (Tables 18.5–18.9, 18.13–18.14) with market share $10\% + 0.25\% \times t$. Value with APV, as in the chapter: unlevered free cash flows and the continuation value at the 10% pretax WACC, plus interest tax shields at the 6.8% debt rate (later tax shields are inside the multiple). Assumption, stated because the statement says "adjusted accordingly": capex is \$5 million in 2011 (replacement only), \$25 million in 2012 (expansion year) and \$8 million from 2013, mirroring the chapter's schedule one year later.

Notation.

- $Q_t = M_t \times s_t$: production (000 units); $S_t = Q_t \times p_t$: sales (\$000); costs 78% of sales; $\tau = 35\%$.
- NWC_t: cash 16% + receivables 19% + inventory 20% − payables 16% of sales = 39% of sales (2010 actual: 29,207).
- $FCF_t = NI_t + (1-\tau)Int_t + Dep_t - \Delta NWC_t - CapEx_t$; $CV_{2015} = 8.5 \times EBITDA_{2015}$.

Step 1 — (18.12) Capacity. Volumes $Q_t = M_t s_t$ (e.g. 2012: $11{,}025 \times 10.5\% = 1{,}157.625$):

	2010	2011	2012	2013	2014	2015
Market share (%)	10.00	10.25	10.50	10.75	11.00	11.25
Production (000 units)	1,000.00	1,076.25	1,157.63	1,244.42	1,337.05	1,435.84

Answer (18.12). Requirements as in the table; production first exceeds 1,100 in 2012, so the expansion is needed in 2012.

Step 2 — (18.13) Interest and tax shields. By Eq. (18.1), interest in year t is 6.8% of debt at the end of $t-1$. Debt is 4,500 through 2011 and 24,500 from the end of 2012: $0.068(4{,}500) = 306$ in 2011 and 2012; $0.068(24{,}500) = 1{,}666$ in 2013–2015. Tax shields = $0.35\times$ interest: 107.1, 107.1, 583.1, 583.1, 583.1.

Answer (18.13). Interest 306, 306, 1,666, 1,666, 1,666; tax shields 107.1, 107.1, 583.1, 583.1, 583.1 ($000).

Step 3 — (18.14) Pro forma income statement. Sales = $Q_t p_t$ (2011: $1{,}076.25 \times 76.51 = 82{,}343.9$); then costs, EBITDA, depreciation (given), interest (Step 2), tax 35%:

$000	2010	2011	2012	2013	2014	2015
Sales	74,889.0	82,343.9	90,341.1	99,055.8	108,555.1	118,916.1
Costs except dep. (78%)	−58,413.0	−64,228.2	−70,466.0	−77,263.5	−84,673.0	−92,754.5
EBITDA	16,476.0	18,115.7	19,875.0	21,792.3	23,882.1	26,161.5
Depreciation	−5,492.0	−5,443.0	−7,398.0	−7,459.0	−7,513.0	−7,561.0
EBIT	10,984.0	12,672.7	12,477.0	14,333.3	16,369.1	18,600.5
Interest	−306.0	−306.0	−306.0	−1,666.0	−1,666.0	−1,666.0
Pretax income	10,678.0	12,366.7	12,171.0	12,667.3	14,703.1	16,934.5
Income tax (35%)	−3,737.3	−4,328.3	−4,259.9	−4,433.5	−5,146.1	−5,927.1
Net income	6,940.7	8,038.3	7,911.2	8,233.7	9,557.0	11,007.4

Answer (18.14). Net income 8,038.3; 7,911.2; 8,233.7; 9,557.0; 11,007.4 for 2011–2015 ($000).

Step 4 — (18.15) Working capital. Each item is its percent of sales (2010 actual):

$000	2010	2011	2012	2013	2014	2015
Cash (16%)	11,982.0	13,175.0	14,454.6	15,848.9	17,368.8	19,026.6
Accounts receivable (19%)	14,229.0	15,645.3	17,164.8	18,820.6	20,625.5	22,594.1
Inventory (20%)	14,978.0	16,468.8	18,068.2	19,811.2	21,711.0	23,783.2
Total current assets	41,189.0	45,289.1	49,687.6	54,480.7	59,705.3	65,403.8
Accounts payable (16%)	11,982.0	13,175.0	14,454.6	15,848.9	17,368.8	19,026.6
Net working capital	29,207.0	32,114.1	35,233.0	38,631.8	42,336.5	46,377.3
Increase in NWC		2,907.1	3,118.9	3,398.8	3,704.7	4,040.8

Answer (18.15). NWC as in the table; increases 2,907.1; 3,118.9; 3,398.8; 3,704.7; 4,040.8 ($000).

Step 5 — (18.20) Free cash flows. Unlevered net income = $NI + (1-0.35)Int$ ($= 0.65\,EBIT$; after-tax interest is 198.9 while debt is 4,500 and 1,082.9 after). E.g. 2012: $8{,}110.1 + 7{,}398 - 3{,}118.9 - 25{,}000 = -12{,}610.8$.

$000	2011	2012	2013	2014	2015
Net income	8,038.3	7,911.2	8,233.7	9,557.0	11,007.4
Plus: after-tax interest	198.9	198.9	1,082.9	1,082.9	1,082.9
Unlevered net income	8,237.2	8,110.1	9,316.6	10,639.9	12,090.3
Plus: depreciation	5,443.0	7,398.0	7,459.0	7,513.0	7,561.0
Less: increase in NWC	−2,907.1	−3,118.9	−3,398.8	−3,704.7	−4,040.8
Less: capital expenditures	−5,000.0	−25,000.0	−8,000.0	−8,000.0	−8,000.0
Free cash flow of firm	5,773.1	−12,610.8	5,376.9	6,448.2	7,610.6

Answer (18.20). FCF = 5,773.1; −12,610.8; 5,376.9; 6,448.2; 7,610.6 ($000).

Step 6 — (18.21) Continuation value. By Eq. (18.7),

$$CV_{2015} = 8.5 \times EBITDA_{2015} = 8.5(26{,}161.5) = 222{,}373.0. \tag{39}$$

Answer (18.21). About $222.4 million at the end of 2015.

Step 7 — (18.22) APV. Discount FCF and CV at 10% ($1.1^5 = 1.61051$) and tax shields at 6.8%:

$$\begin{aligned} PV(FCF) &= \frac{5{,}773.1}{1.1} + \frac{-12{,}610.8}{1.1^2} + \frac{5{,}376.9}{1.1^3} + \frac{6{,}448.2}{1.1^4} + \frac{7{,}610.6}{1.1^5} \\ &= 5{,}248.3 - 10{,}422.2 + 4{,}039.7 + 4{,}404.2 + 4{,}725.6 = 7{,}995.6, \\ PV(CV) &= 222{,}373.0/1.61051 = 138{,}076.2, \\ PV(ITS) &= \frac{107.1}{1.068} + \frac{107.1}{1.068^2} + \frac{583.1}{1.068^3} + \frac{583.1}{1.068^4} + \frac{583.1}{1.068^5} \\ &= 100.3 + 93.9 + 478.7 + 448.2 + 419.6 = 1{,}540.7, \\ V_{2010} &= 7{,}995.6 + 138{,}076.2 + 1{,}540.7 = 147{,}612.5. \end{aligned} \tag{40}$$

Answer (18.22). KMS is worth about $147.6 million under the 0.25% scenario, versus $201.0 million in the chapter's 1% scenario (Table 18.14) — $53.4 million less (still $42.5 million less if the chapter's value is redone at 8.5×: $190.1 million).

Intuition. The value lives in the continuation value (93.5% of the total), and the continuation value lives in 2015 EBITDA. Slower share growth cuts 2015 EBITDA from 34,880 to 26,162, and with it most of the value; the five-year cash flows barely matter. That is the chapter's lesson: growth is valuable only through the cash flows it adds, and the forecast's horizon multiple must be stress-tested.

Common mistakes.

- Subtracting the level of NWC instead of its *increase* (Berk's "Common Mistake", p. 553).
- Discounting the tax shields at 10%, or adding them twice (net income already contains them; start from *unlevered* income).
- Charging the new bond's interest in 2012: under Eq. (18.1) interest starts the year after the borrowing.

Ref: Berk §18.3–18.5, pp. 542–556; Tables 18.5–18.9, 18.13–18.14.